

Topics -

Input / Output

1) Input / Output -

- a) Peripheral Devices
- b) I/O interface
- c) I/O ports

2) Interrupts -

- a) Interrupt Hardware
- b) Types of interrupts
- c) and Exceptions

3) Modes of Data Transfer -

- a) Programmed I/O
- b) Interrupt initiated I/O
- c) Direct Memory Access
- d) I/O channels and Processors

4) Serial Communication -

- a) Synchronous & Asynchronous Commⁿ
- b) Standard Commⁿ Interfaces

Peripheral Devices

Peripheral devices are the computer devices that are connected to the computer externally such as printer, scanner, keyboard, mouse and external modem. It can be internal such as CD-ROM or internal modem.

Classification -

- 1) Input devices - Are the type of computer devices that are used to provide control signals to the computer. Eg - Keyboard, mouse etc.
- 2) Output devices - Are devices that are used to display the results. Eg - Printer, monitor etc.
- 3) Storage devices - A device that is used to store the information such as hard-disk drive, flash drive etc.

Advantages -

- 1) It is helpful for taking input very easily.
- 2) Provide a specific output.
- 3) has a storage device for storing info.
- 4) Improves the efficiency of system.

Interface (I/O Interface)

b) 3, 5, 11, 8, 4, 1, 12, 7, 2, 6, 10

Values

- 1) 3
- 2) 5

Insertion



Rotation

3) 11

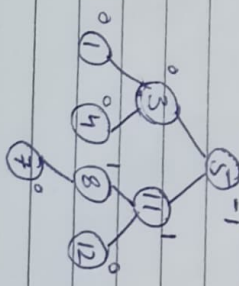
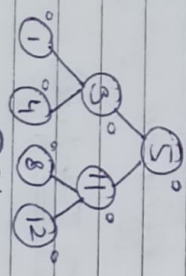
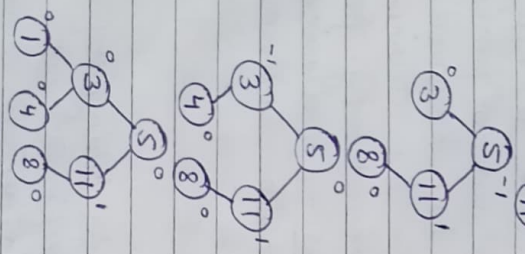
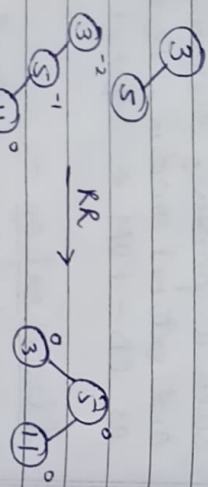
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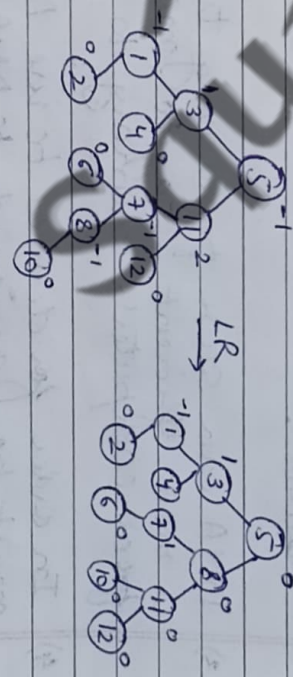
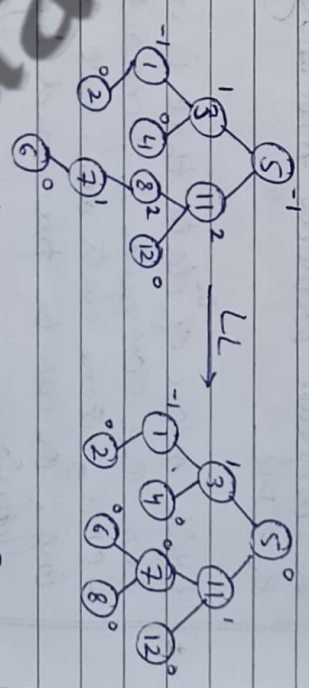
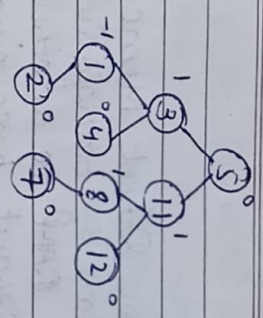
6) 1

7) 12

8) 7



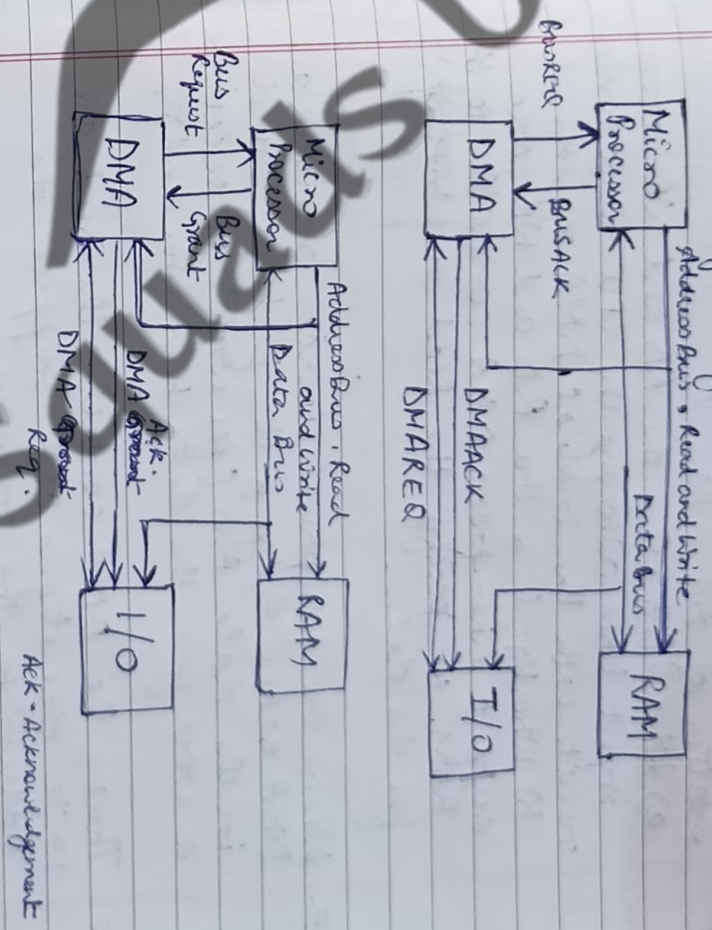
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DMA

- 1) DMA stands for Direct Memory Access and is a method of transferring data from the computer's RAM to another part of the computer without processing it using the CPU.
- 2) While most of the data that is input or output from our computer is processed by the CPU, some data does not require processing, as can be processed by another device.
- 3) DMA can save processing time and is a more efficient way to move data from computer's memory to other devices.
- 4) In order for devices to use DMA, they must be assigned to DMA channel. Each type of port on a computer has a set of DMA channels that can be assigned to each connected device.
- 5) Ex Eg - A PCI controller and hard drive controllers each have their own set of DMA channels.

Block Diagram for DMA :-



Working

- 1) When the peripheral device sends a DMA request, the DMA controller activates the BR (Bus Request) line, informing the CPU to relinquish the bus.
- 2) The CPU responds with its BG line, informing the DMA that its bus is available.
- 3) The DMA then puts the current value of its address register into the address bus, initiates the RD or WR signal and sends a DMA acknowledge to the peripheral device.

4. The direction of transfer depends on the status of BQ line.
 - a) When $BQ = 0$, the RD and WR are input lines allowing the CPU to communicate with the internal DMA register.
 - b) When $BQ = 1$, the RD and WR are output lines from DMA controller to the CPU to specify the Read or Write opⁿ for the data.
 5. When the peripheral device receives a DMA acknowledge, it puts a word in the data bus (for write) or receives a word from the data bus (for read).
 6. Thus, the DMA controls the read or write operations and supplies the address for the memory.
 7. The peripheral unit can communicate with memory through the data bus for direct transfer b/w two units while the CPU is momentarily disabled.
- ⇒ Reason for bi-directional RD and WR control line → Because the microprocessor gets (read) the data from the memory and write data to the memory (address both).

⇒ DMA transfer in a computer system -

